

ALFAAL02985

## tert-Butyldimethylchlorosilane

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | tert-Butyldimethylchlorosilane<br>tert-Butyldimethylchlorosilane                                                                                                                                                                                                   |
| Cat No. :                     | L02985                                                                                                                                                                                                                                                             |
| Synonyms                      | TBDMCl; tert-Butyldimethylsilyl chloride                                                                                                                                                                                                                           |
| CAS No                        | 18162-48-6                                                                                                                                                                                                                                                         |
| Molecular Formula             | C <sub>6</sub> H <sub>15</sub> Cl Si                                                                                                                                                                                                                               |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                             |                     |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------|
| Physical State<br>Solid                                                                                                                                     | Appearance<br>White | Odor<br>No information available |
| <b>Emergency Overview</b><br>Flammable solid. Causes severe skin burns and eye damage. Toxic to aquatic life with long lasting effects. Moisture sensitive. |                     |                                  |

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Flammable solids.                 | Category 2   |
| Skin Corrosion/Irritation         | Category 1 A |
| Serious Eye Damage/Eye Irritation | Category 1   |
| Chronic aquatic toxicity          | Category 2   |

#### Label Elements



# SAFETY DATA SHEET

## tert-Butyldimethylchlorosilane

**Signal Word****Danger****Hazard Statements**

H228 - Flammable solid

H314 - Causes severe skin burns and eye damage

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Highly flammable. Vapors may cause flash fire or explosion. Combustible material. Water reactive.

**Health Hazards**

Corrosive. Causes skin and eye burns.

**Environmental hazards**

Toxic to aquatic life with long lasting effects. Is not likely mobile in the environment. Hydrolyses.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                  | CAS No     | Weight % |
|--------------------------------------------|------------|----------|
| Silane, chloro(1,1-dimethylethyl)dimethyl- | 18162-48-6 | <= 100   |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Keep eye wide open while rinsing.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask

# SAFETY DATA SHEET

## tert-Butyldimethylchlorosilane

equipped with a one-way valve or other proper respiratory medical device.

### Ingestion

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

### Most important symptoms and effects

Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Notes to Physician

Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

Do not use halon type extinguisher, Do not use water jetstream.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

### Personal Precautions

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water.

### Storage

Flammables area. Purge open drums with nitrogen before resealing. Corrosives area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place.

## tert-Butyldimethylchlorosilane

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## tert-Butyldimethylchlorosilane

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                             |                                          |
|------------------------------------------------|-----------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                       |                                          |
| <b>Physical State</b>                          | Solid                       |                                          |
| <b>Odor</b>                                    | No information available    |                                          |
| <b>Odor Threshold</b>                          | No data available           |                                          |
| <b>pH</b>                                      | No information available    | decomp                                   |
| <b>Melting Point/Range</b>                     | 85 - 91 °C / 185 - 195.8 °F |                                          |
| <b>Softening Point</b>                         | No data available           |                                          |
| <b>Boiling Point/Range</b>                     | 125 °C / 257 °F             | @ 760 mmHg                               |
| <b>Flash Point</b>                             | 22 °C / 71.6 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable              | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available    |                                          |
| <b>Explosion Limits</b>                        | No data available           |                                          |
| <b>Vapor Pressure</b>                          | 12 hPa @ 20 °C              |                                          |
| <b>Vapor Density</b>                           | Not applicable              | Solid                                    |
| <b>Specific Gravity / Density</b>              | 0.870                       |                                          |
| <b>Bulk Density</b>                            | No data available           |                                          |
| <b>Water Solubility</b>                        | hydrolyses                  |                                          |
| <b>Solubility in other solvents</b>            | No information available    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                             |                                          |
| <b>Autoignition Temperature</b>                | 405 °C / 761 °F             |                                          |
| <b>Decomposition Temperature</b>               | No data available           |                                          |
| <b>Viscosity</b>                               | Not applicable              | Solid                                    |
| <b>Explosive Properties</b>                    | No information available    |                                          |
| <b>Oxidizing Properties</b>                    | No information available    |                                          |
| <b>Molecular Formula</b>                       | C6 H15 Cl Si                |                                          |
| <b>Molecular Weight</b>                        | 150.73                      |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Moisture sensitive.                                                                                                                                                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                                        |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                             |
| <b>Conditions to Avoid</b>              | Avoid dust formation. Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moisture. |
| <b>Materials to avoid</b>               | Bases. Water. Strong oxidizing agents. Alcohols. Amines. Metals.                                                                                                     |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Chlorine. Silicon dioxide. Hydrogen chloride gas.                                                           |

## SECTION 11. TOXICOLOGICAL INFORMATION

**Product Information** The toxicological properties have not been fully investigated

## (a) acute toxicity;

| Component                                  | LD50 Oral         | LD50 Dermal | LC50 Inhalation |
|--------------------------------------------|-------------------|-------------|-----------------|
| Silane, chloro(1,1-dimethylethyl)dimethyl- | >2000 mg/kg (Rat) |             |                 |

## (b) skin corrosion/irritation;

Category 1 A

**SAFETY DATA SHEET**  
**tert-Butyldimethylchlorosilane**

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**SECTION 12. ECOLOGICAL INFORMATION**

**Ecotoxicity effects**

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and Degradability**

**Persistence**

**Degradability**

**Degradation in sewage treatment plant**

No information available

Persistence is unlikely, based on information available.

Decomposes in contact with water.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Decomposes in contact with water.

**Bioaccumulative Potential**

Product does not bioaccumulate due to reaction with water

**Mobility in soil**

Hydrolyses Is not likely mobile in the environment

**Endocrine Disruptor Information**

**Persistent Organic Pollutant**

**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS**

## tert-Butyldimethylchlorosilane

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN2921  
 Proper Shipping Name Corrosive solid, flammable, n.o.s.  
 Technical Shipping Name tert-Butylchlorodimethylsilane  
 Hazard Class 8  
 Subsidiary Hazard Class 4.1  
 Packing Group I

**IMDG/IMO**

UN-No UN2921  
 Proper Shipping Name Corrosive solid, flammable, n.o.s.  
 Technical Shipping Name tert-Butylchlorodimethylsilane  
 Hazard Class 8  
 Subsidiary Hazard Class 4.1  
 Packing Group I

**IATA**

UN-No UN2921  
 Proper Shipping Name Corrosive solid, flammable, n.o.s.  
 Technical Shipping Name tert-Butylchlorodimethylsilane  
 Hazard Class 8  
 Subsidiary Hazard Class 4.1  
 Packing Group I

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                  | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Silane, chloro(1,1-dimethylethyl)dimethyl- | -                                                   | -                                       | X    | X     | 242-042-4 | X    | -   | X     | X    | X    | X    | 97-3-163 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 03-Jul-2014  
**Revision Date** 07-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**